THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 8: **Nature of Matter: Elements, Compounds and Mixtures** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of the following pairs contains only elements? 1

(a) $NaCl$ and Fe **(b)** H_2O and O_2
(c) O_2 and Fe **(d)** CO_2 and N_2

2. Which of the following is the odd one out? 1

(a) sulphur **(b)** mercury **(c)** iron **(d)** copper

3. A bell is made of bronze because it is: 1

(a) liquid **(b)** soft
(c) brittle **(d)** sonorous and hard

4. Which of the following pairs contains only compounds? 1

(a) NH_3 and CH_4 **(b)** Fe and CO_2
(c) O_3 and CO_2 **(d)** N_2 and CH_4

5. Which of the following is the odd one out? 1

(a) steel **(b)** bronze **(c)** brass **(d)** copper

6. Electric wires are made of copper because copper is: 1

(a) light and non-conducting **(b)** brittle
(c) a good conductor and ductile **(d)** magnetic

7. A student heated a mixture of iron and sulphur strongly and got a black solid that is not magnetic. The black solid is a: 1

(a) element **(b)** mixture **(c)** alloy **(d)** compound

8. Which element is both lustrous and brittle? 1

(a) iron **(b)** silicon **(c)** gold **(d)** copper

9. The cooking utensils are made of metals because they are: 1

(a) poor conductors **(b)** brittle
(c) good conductors of heat **(d)** lighter than water

10. The ratio of hydrogen to oxygen by mass in water is: 1

(a) 2 : 1 **(b)** 1 : 2 **(c)** 1 : 8 **(d)** 8 : 1

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why is water called a compound and not a mixture? Give two reasons. 2

12. How many atoms are present in one molecule of C_2H_5OH? 2

13. Why is the formula of water not written as HO? 2

14. Why does an iron-sulphur mixture behave differently from iron sulphide? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A student mixed iron filings and sulphur and heated them. Write three observations and the inference. 3

16. Give three reasons why gold and silver are used in jewellery. 3

17. Why are metals used for making bells, utensils and wires? Give properties. 3

OR

How can you distinguish a mixture of iron and sulphur from iron sulphide? Give three tests. 3

18. Explain the importance of noble gases and give two uses. 3

OR

A substance X is a shiny solid that conducts electricity and can be drawn into wires. State whether X is a metal and give another property. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the properties of the iron-sulphur mixture and iron sulphide, and conclude what is seen about mixtures and compounds. 5

OR

Discuss the use of metals and non-metals in the human body, daily life and industry with five examples. 5

20. Differentiate between elements, compounds and mixtures giving the definition, composition, separation and three examples each. 5

OR

A student heated mercuric oxide and obtained a silvery liquid and a gas that rekindled a glowing splinter. Identify and explain what the experiment shows about compounds. 5

END OF QUESTION PAPER — Total: 40 Marks

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